

UCS758: System Provisioning and Configuration Management

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Syllabus

Understanding Containers: Transporting goods analogy and its problems, Containerization platform, images and runtime, comparison with virtual machine, chroot system call, FreeBSD Jails, Linux containers (LXC), Docker.

Introduction to Containerization: Docker architecture, different environments (Dev, QA and Prod), overcoming issues with different environments, virtual machine for dev/deployments, containers for dev/deployments, advantages and drawbacks of containerization.

Orchestration Tools: Orchestration: its definition and need, Docker swarm and Kubernetes, AWS (ECS and EKS), Kubernetes on cloud, monitoring containers and its process.

Introduction to Provisioning: Basic and software definition, provisioning concepts, reason for exclusive provisioning, configuration management definition and tools, difference between provisioning and configuration management, provisioning tools, test machines for provisioning, deployment and its relationship with provisioning.

On Premise Provisioning: Understanding and Defining On Premise, On Premise provisioning infrastructure, Templating, server templating and its challenges.

Provisioning on Cloud: defining cloud provisioning, types of cloud provisioning, life-cycle of provisioning on cloud, On Premise cloud mitigation strategies, network security enablement from On Premises to cloud, micro-services management in cloud.

Provisioning and Configuration Management: State of tools in provisioning and configuration, definition and need for configuration management, its benefits and drawbacks in DevOps, need for monitoring in DevOps, reasons for using provisioning and configuration tools, automation, preventing errors and tracking changes, examples of tools and their capabilities.

Laboratory Work: System Provisioning: Automation of infrastructure, AWS configuration for Terraform, create IAM User, security group, spinning up with EC2 instance, variables, resources, modules, state management, VPC, IAM policy, S3 bucket and its variables.

Containers Lab: Playing with Vagrant and understanding its file, Docker machine, Dockerfile, Docker extras, DTR, Docker compose and swarm, Kubernetes -Minikube, deploying Pods and services on Minikube.

Course Learning Objectives (CLO)

After the completion of the course the student will be able:

1. Understand the concept of Virtualization and Containerization.
2. Familiarize Orchestration and System provisioning.
3. Analyze and demonstrate the infrastructure automation and state management in the cloud environment.
4. Understand and demonstrate the need for configuration management